Unit 3- OR

1. Question

The critical path duration of a project is 10 days with a standard deviation of 4 days. Compute the probability of completing the project in 46 days or less.

Hint: P(0<z<4)=0.9999.

1. **Given:**
   - Critical path duration (μ) = 10 days
   - Standard deviation (σ) = 4 days
   - Target duration (X) = 46 days
2. **Z-score formula:**

$$z = \frac{X - \mu}{\sigma}$$

3. **Substitute values:**

$$z = \frac{46 - 10}{4} = \frac{36}{4} = 9$$

4. **Conclusion:**
   Since $z = 9$ is much greater than 4, and $P(0 < z < 4) = 0.9999$, the probability for $z = 9$ is approximately **1 (100%)**.

The company managing a project where Task A has the following estimates:

Optimistic time (O): 7 days

Most likely time (M): 5 days

Pessimistic time (P): 6 days

Using the PERT formula,

a) Calculate the average duration of Task A.

b) Task X has the following estimates: O: 3 days, M: 4 days, P: 9 days

Calculate the mean and standard deviation for Task X using PERT.

(a) Average duration of Task A using the PERT formula:

The PERT formula for the average duration is:

$$\text{Average Duration} = \frac{O + 4M + P}{6}$$

Substitute the values for Task A:

$$\text{Average Duration} = \frac{7 + 4(5) + 6}{6} = \frac{7 + 20 + 6}{6} = \frac{33}{6} = 5.5\,\text{days}$$

Answer: The average duration of Task A is **5.5 days**.

b)

1. **Mean (Average Duration):**

$$\text{Mean} = \frac{O + 4M + P}{6}$$

Substitute the values for Task X:

$$\text{Mean} = \frac{3 + 4(4) + 9}{6} = \frac{3 + 16 + 9}{6} = \frac{28}{6} = 4.67\,\text{days}$$

2. **Standard Deviation (σ):**

   The formula for the standard deviation in PERT is:

$$\sigma = \frac{P - O}{6}$$

Substitute the values for Task X:

$$\sigma = \frac{9 - 3}{6} = \frac{6}{6} = 1\,\text{day}$$

Answer:

- The mean of Task X is **4.67 days**.
- The standard deviation of Task X is **1 day**.

A manufacturing company places a semi-annual order of 48,000 units at a price of $50 per unit. Its carrying cost is 5% and the order cost is $10 per order.

a) Calculate the economic order quantity of the manufacturing company.

b) Determine the number of orders that need to be placed.

c) Compute the annual requirements of the company.

(a) Economic Order Quantity (EOQ):

The EOQ formula is:

$$EOQ = \sqrt{\frac{2DS}{H}}$$

1. Annual demand (D) = $48,000 \text{ units (semi-annual)} \times 2 = 96,000 \text{ units per year}$
2. Substitute the values into the formula:

$$EOQ = \sqrt{\frac{2 \times 96,000 \times 10}{2.5}}$$

$$EOQ = \sqrt{\frac{1,920,000}{2.5}} = \sqrt{768,000} \approx 876.66$$

Answer: The economic order quantity (EOQ) is approximately **877 units**.

Steps:

1. The semi-annual demand is **48,000 units**, so the **annual demand (D)** is:

$$D = 48,000 \times 2 = 96,000 \text{ units}$$

2. The carrying cost (H) is **5% of the unit price**:

$$H = 0.05 \times 50 = 2.5 \text{ per unit per year}$$

3. Substitute $D = 96,000$, $S = 10$, and $H = 2.5$ into the EOQ formula:

$$EOQ = \sqrt{\frac{2 \times 96,000 \times 10}{2.5}}$$

$$EOQ = \sqrt{\frac{1,920,000}{2.5}} = \sqrt{768,000} \approx 876.66$$

4. Round the result to get:

$$EOQ \approx 877 \text{ units}$$

Conclusion: The optimal order quantity is **877 units**.

(b) Number of Orders to be Placed:

The number of orders is calculated as:

$$\text{Number of Orders} = \frac{D}{EOQ}$$

Substitute the values:

$$\text{Number of Orders} = \frac{96,000}{877} \approx 109.4$$

Answer: The company needs to place approximately **109 orders** per year.

(c) Annual Requirements of the Company:

The annual requirement is the total demand for the year, which is already calculated:

Annual Requirements=96,000 units\text{Annual Requirements} = 96,000 \, \text{units}Annual Requirements=96,000units

Answer: The annual requirement of the company is **96,000 units**.